Esandu Meth Obadaarachchi

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About

BSc (Hons) Artificial Intelligence and Data Science graduate with First Class Honours and two years of industry experience at SLT Telecom, building production AI systems. IEEE-published researcher. Hands-on experience across Generative AI workflows and computer vision.

Key Skills and Qualifications

Programming and ML: Python, scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, BERT, DistilBERT, LSTM, CNN, YOLO, MAML, CycleGAN

Development: Backend: FastAPI, Node.js | Frontend: React JS, Next.js | Mobile: Flutter, React Native | Languages: Python, JavaScript, Dart, Java, SQL

Libraries and Tools: MLflow, Docker, LangChain, LangChain.js, Pinecone, pandas, NumPy, OpenCV, Git, Firebase, MongoDB, InfluxDB, N8n, Pydantic, Power BI, Hugging Face Transformers

Cloud and MLOps: AWS, Azure ML, GitHub Actions, CI/CD pipelines, Gemini API, OpenAI API, Claude API

Areas: Computer Vision, NLP, Deep Learning, Anomaly Detection, RAG Pipelines, Time-Series Forecasting, Medical Image Processing, Agentic AI, Model Explainability, Prompt Engineering, Genetic Algorithms, Evolutionary Algorithms, LLM Fine-tuning

Experience

Freelance AI Engineer *June 2025 – Present*

SLT Telecom PLC, Head Office (Power and AC Division + Digital Platform Section)

- Worked across the power tech domain, developing and deploying a RAG-based agentic AI workflow within SLT PowerProx that guides field engineers through fault troubleshooting, step-by-step procedures and in-app navigation.
- Led SLT PowerProx and SLT PowerZenith from development to production as Project Lead Developer and Project Manager, managing a team of junior developers across task assignment, code review and system integration.
- Built an AI-powered anomaly detection pipeline for SLT PowerZenith using XGBoost with lag features to flag irregular energy consumption patterns.

AI Engineer Intern *July 2024 – June 2025*

SLT Telecom PLC, Head Office (Power and AC Division + Digital Platform Section)

- Built SLT TalentGrid, an ATS tool that ranks intern CVs against recruiter prompts using vector similarity, integrated with Gmail API.
- Integrated an AI-powered quotation analyser into SLT PowerProx using LLM APIs to automate cost estimation.

Key Achievements [SLT Telecom PLC]

- SLT PowerProx: BMPC 2025 Telecommunications Category Winner and Digital Literacy Cluster (Twenty Outstanding Companies); NPME 2025 Silver Award for Best Managed Project in the Telecom Sector.
- SLT PowerZenith: BMPC 2026 Digital Literacy Cluster Award Winner and Among the Forty Outstanding Companies.

Private Tutor *2022 – Present*

Self-Employed

- Teaching Cambridge and Edexcel O Level Chemistry, Physics and Mathematics.
- Teaching Cambridge and Edexcel A Level Mathematics.

Projects

Lune AI: Project/Task Management + Agentic RAG Second-Brain (FastAPI + LangGraph) - luneai.site

- Built the second-brain assistant as a multi-tenant agentic RAG system, modelling the retrieval loop as a LangGraph state machine (query rewrite, vector search, cross-encoder rerank, relevance grading, retry) with a grounded check to cut hallucination.
- Two-stage retrieval (Voyage embeddings, cross-encoder rerank) over Pinecone, feeding a Claude ReAct agent with tool-calling access to tasks and knowledge.

Tools Used: FastAPI, LangChain, LangGraph, Claude, Voyage AI, Pinecone, Firebase/Firestore, Docker, AWS ECS Fargate, Next.js

Few-Shot Cross-Modal Medical Image Translation *[Individual Final Year Project Research]*

- Developing a CT to MRI and MRI to CT medical image synthesis system using Meta-Learning (MAML) and CycleGAN for cross-modal translation.
- Applying Coordinate Attention mechanisms to preserve spatial features in generated medical images, improving diagnostic utility.

Tools Used: Python, TensorFlow 2.10, Keras, MAML, CycleGAN, Coordinate Attention, Google Colab, Vast AI

GreenGuard: AI-Powered Tea Leaf Disease Detection *[IEEE Published]*

- Published at the 2024 6th International Conference on Advancements in Computing (ICAC), IEEE Xplore. Paper: ieeexplore.ieee.org/document/10851118
- Developed a CNN and transfer learning model for real-time tea leaf disease classification from mobile camera input, achieving accurate multi-class detection.
- Built a React Native frontend with Node.js backend for on-device inference and treatment guidance delivery.

Tools Used: Python, TensorFlow, Keras, CNN, Transfer Learning, React Native, Node.js

AI-Augmented Hotel Management Platform *[Freelance - The Odon Pvt Ltd]*

- Developed a full-stack hotel management application covering booking management, inventory tracking, profit and revenue calculation and staff management using Flutter for Android and web along with CI/CD pipelines.
- Integrated Gemini API for AI-powered expense entry via image and text input, reducing manual data entry overhead and for business financial analysis.

Tools Used: Flutter, MongoDB, Node.js, Gemini API

NLP-Based Public Opinion Analysis System

- Built an end-to-end NLP pipeline for topic modelling, sentiment analysis and named entity extraction from social media.
- Trained and evaluated BERT, DistilBERT, Word2Vec, LSTM, CNN and SVM models; applied LDA for topic modelling across Reddit data via PRAW API.

Tools Used: Python, BERT, DistilBERT, Word2Vec, LSTM, CNN, SVM, LDA, PRAW API, scikit-learn

SmartSmile, Edge AI Dental Diagnostic App (Flutter, YOLOv10, TensorFlow Lite)

- Designed the AWS backend: S3, Lambda, SageMaker for automated retraining, Elastic Beanstalk, EventBridge and CloudWatch, with a human-in-the-loop annotation pipeline via Label Studio for low-confidence predictions.

Tools Used: Flutter, Dart, TensorFlow Lite, YOLOv10, OpenCV, AWS (S3, Lambda, SageMaker, Elastic Beanstalk, EventBridge, CloudWatch, IAM), Label Studio, Docker

Education

Robert Gordon University, Aberdeen, UK (affiliated with Informatics Institute of Technology, Sri Lanka)
BSc (Hons) Artificial Intelligence and Data Science | First Class Honours | 2022 – 2026

Lyceum International School Nugegoda, Sri Lanka
Cambridge A Levels, Mathematics Stream - 3A*s | 2020 – 2022
Cambridge O Levels, Science Stream - 4A*, 2As, 1C | 2018 – 2020

Certifications

- Neural Networks and Deep Learning - Coursera: DeepLearning.AI
- Multimodal RAG with GPT - Coursera: Packt
- Generative AI Advanced Fine-Tuning for LLMs - Coursera: IBM

References

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